

Universal Confounder Stress Test for Bio-AI

A preregistered, three-domain evaluation of whether a model-agnostic audit score can detect validation optimism.

LOCKED OUTCOME: NEGATIVE

The composite did not satisfy the all-required success gate. Leave-one-domain-out sensitivity was 0.571, and two leave-one-dimension-out variants outperformed the full score. A stricter protocol-compliance audit also found that four nominal calculator passes were only partially implemented and therefore cannot be treated as demonstrated.

Round completed: 21 September 2026

Protocol SHA-256: dcb430c4306ef81b6cc7fdc96b1a2763b960e915c84e4cba11d951f7ab6ae849

Open-data computational research. No wet-lab work. This report is a negative-result technical dossier, not a success paper or a validated medical-device claim.

Executive decision

The Round 1 question was whether one fixed audit score, built from model-agnostic diagnostics, could identify configurations whose naive validation overstated rigorous generalization across clinical tabular data, multi-study transcriptomics, and longitudinal digital biomarkers. The answer is **not yet**. The score correlated with measured optimism (Spearman ρ 0.770), and material optimism appeared in breast pCR and Parkinson telemonitoring. However, correlation did not translate into a stable universal detector. Cross-domain sensitivity missed the locked 0.80 threshold, and removing prevalence or calibration improved AUROC.

Operational decision: Do not use this score as an automated release gate, certification tool, or safety claim. It can be retained as an exploratory evidence dossier that forces teams to inspect split design, unit overlap, cohort shift, calibration, and subgroup instability. Human review remains mandatory.

What is solid

- Real open datasets were used, with 99,343 eligible diabetes encounters, five breast cancer studies totaling 741 endpoint-labeled samples, and 5,875 longitudinal Parkinson observations from 42 subjects.
- Naive and group/study-aware evaluations were executed with train-only preprocessing.
- The direction and magnitude of optimism were large for Parkinson models and moderate for pooled breast transcriptomics, while diabetes readmission showed little split-induced inflation.
- The frozen protocol, code, processed result tables, diagnostics, environment, and checksums are preserved.

What is not established

- A transportable threshold that detects material optimism in a new biomedical domain.
- Incremental value from all six implemented score dimensions.
- All inferential and negative-control details specified in the locked protocol.
- End-to-end reproducibility from source archives in a clean environment.

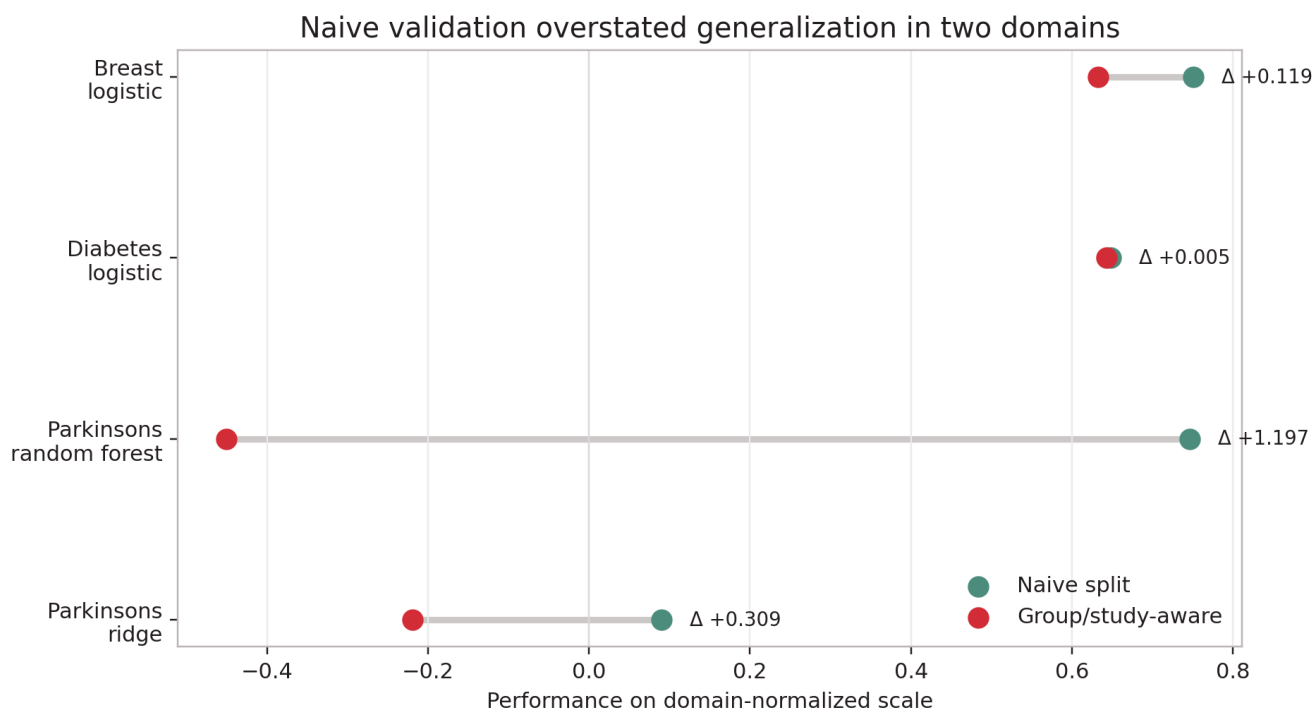


Figure 1. Mean performance under naive and rigorous splits. For classification, performance is AUROC. For Parkinson regression, performance is $1 - \text{RMSE}/\text{SD}(\text{test})$, so values below zero indicate worse-than-mean prediction on held-out subjects. Deltas are naive minus rigorous.}}

Question, protocol, and data

The scientific hypothesis was frozen before outcome modeling: a composite of dependence, feature shift, missingness, prevalence sensitivity, calibration drift, subgroup instability, and cohort/group effects would rank and detect validation optimism. Material optimism was fixed at $\Delta \geq 0.05$ on a domain-normalized performance scale. All six success criteria were required. Failure of any criterion meant a negative Round 1 result.

Domain	Independent unit	Endpoint	Naive design	Rigorous design
Diabetes 130-US Hospitals	Patient	30-day readmission	Random encounter rows	Patient-group split
curatedBreastData	Study / patient	Pathologic complete response	Pooled random split	Leave-one-study-out
Parkinson Telemonitoring	Subject	Total UPDRS	Random observation rows	Leave-subject groups out

Data and preprocessing

For diabetes, records with death or hospice discharge dispositions were excluded and identifiers, outcome fields, encounter order, and discharge disposition were not predictors. Numeric values were median-imputed and standardized; categorical values were mode-imputed and one-hot encoded inside training folds. The final analytic set contained 99,343 encounters from 69,990 patients, with 11,314 30-day readmissions.

For breast pCR, five curated studies were exported from the CC-BY archive. Features were collapsed to gene symbols by median, intersected to 10,557 common genes, imputed and scaled within training data, and reduced to the 500 highest univariate F-score features within each training fold. Endpoint-labeled cohort sizes were 114, 261, 129, 122, and 115.

For Parkinson telemonitoring, subject ID and motor UPDRS were excluded; ridge and random-forest regressors predicted total UPDRS. The normalized score was $1 - \text{RMSE}/\text{SD}(\text{test})$. The held-subject distribution shift caused several negative scores. This is mathematically legitimate but makes the optimism scale unbounded above, an important design weakness for a purportedly universal audit.

Primary results

Domain / model	Naive	Rigorous	Optimism Δ	Interpretation
Breast / logistic	0.7508	0.6323	+0.1186	Material pooled-study inflation
Diabetes / logistic	0.6479	0.6424	+0.0054	Little split-induced inflation
Parkinson / random forest	0.7462	-0.4506	+1.1968	Severe repeated-subject inflation
Parkinson / ridge	0.0898	-0.2190	+0.3088	Material repeated-subject inflation

These effects should not be pooled as if their scales were exchangeable. Breast and diabetes use AUROC, while Parkinson uses a normalized regression skill score. Round 1 treated $\Delta \geq 0.05$ as common, but Parkinson's score can become strongly negative when held-subject RMSE exceeds the test-set standard deviation. Round 2 must test scale harmonization rather than assume it.

Association was strong, but cross-domain detection was unstable
Spearman $\rho = 0.770$; configuration-bootstrap 95% CI 0.444 to 1.000

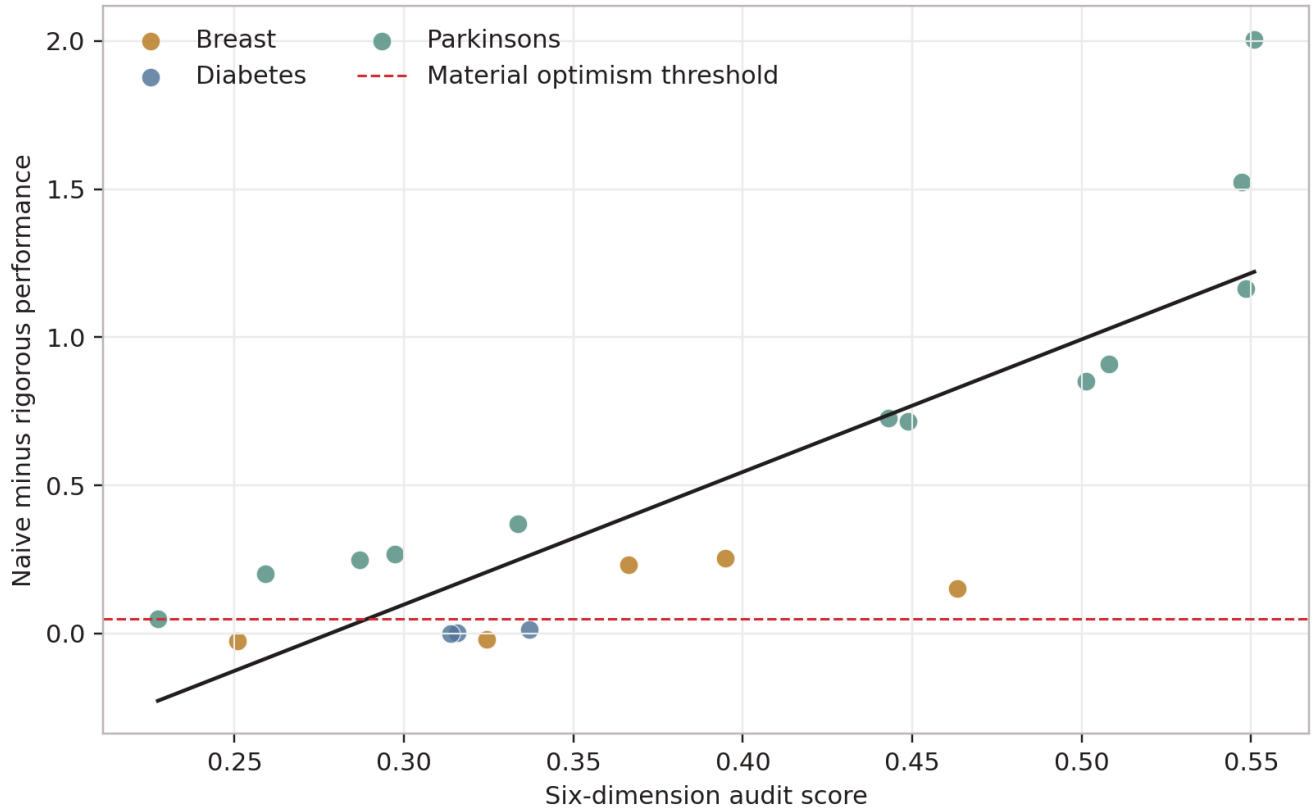


Figure 2. Configuration-level association. The calculator interval is descriptive because the locked independent-unit bootstrap was not implemented.

Detector performance and ablation

The full audit score achieved AUROC 0.845 for classifying configurations with optimism $\Delta \geq 0.05$. This appears useful in isolation, but it failed the stronger preregistered claim. Leave-one-domain-out sensitivity was 0.571 and specificity 0.833. A detector that misses roughly 43% of materially optimistic configurations cannot serve as a safety gate.

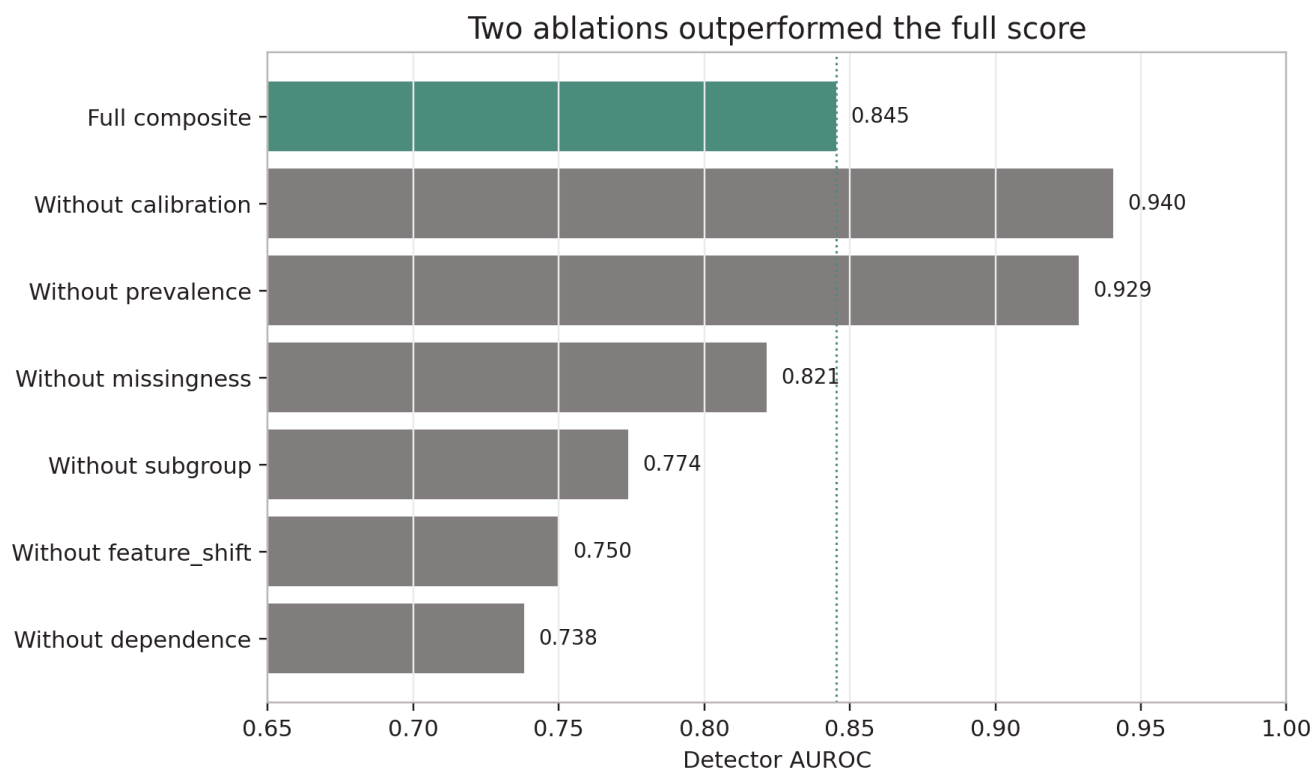


Figure 3. Full composite and leave-one-dimension-out AUROCs. Removing prevalence or calibration increased AUROC, failing G4.

The ablation result suggests that equal weighting is not harmless. In these data, prevalence and calibration diagnostics carried domain-specific structure that diluted transport. That is a mechanistic clue, not permission to tune the score on Round 1 labels. Any simplified score must be frozen as a new Round 2 model and evaluated prospectively.

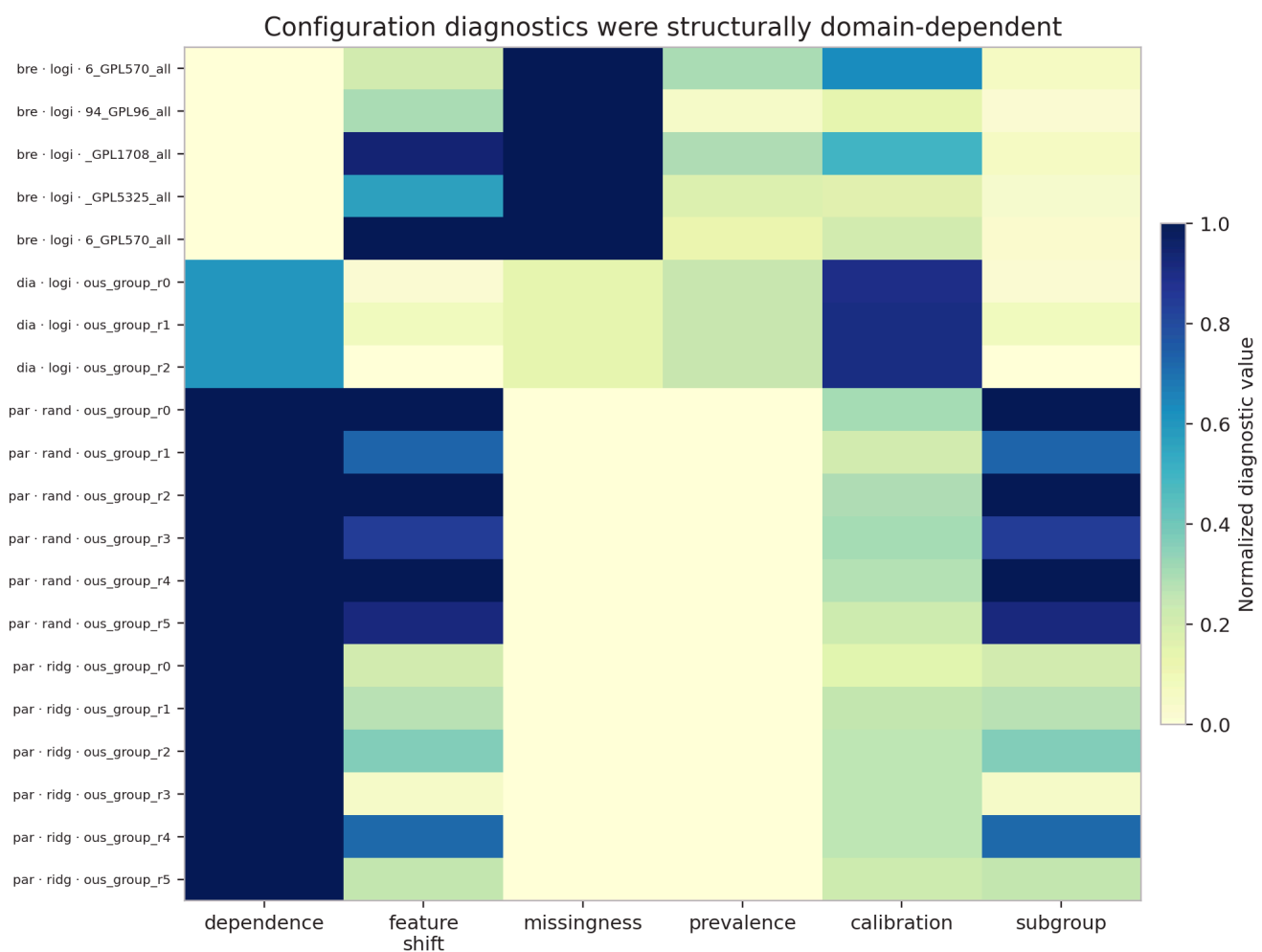


Figure 4. Six diagnostics by configuration. Strong domain blocks help explain poor threshold transport.

Locked gates and protocol-compliance audit

The original calculator marked G1, G2, G5, and G6 as passes, with G3 and G4 failing. A line-by-line audit against the frozen wording found that several passes used narrower computations than specified. Because this program preserves negative and incomplete evidence, the report carries both the calculator snapshot and a conservative adjudication.

Protocol-compliance adjudication

	G1	G2	G3	G4	G5	G6
Calculator	Pass	Pass	Fail	Fail	Pass	Pass
Adjudicated	Not demonstrated	Not demonstrated	Fail	Fail	Not demonstrated	Not demonstrated

The locked all-required gate remains negative under either view. Adjudication is stricter because several required analyses were not actually implemented.

Figure 5. Calculator flags versus conservative protocol-compliance adjudication.

Gate	Calculator	Adjudicated	Reason
G1 correlation + bootstrap	Pass	Not demonstrated	$\rho = 0.770$, but bootstrap resampled configurations, not raw independent units.
G2 two domains + Holm p	Pass	Not demonstrated	Two point estimates exceeded 0.05; paired unit-level tests and Holm adjustment were absent.
G3 leave-domain detection	Fail	Fail	Sensitivity $0.571 < 0.80$; specificity 0.833.
G4 ablation superiority	Fail	Fail	Prevalence and calibration ablations outperformed the full composite.
G5 controls	Pass	Not demonstrated	Overlap and duplicate tests passed, but not all frozen controls were run as written.
G6 clean rerun	Pass	Not demonstrated	Only the gate calculator's semantic JSON was rerun, not the full data-to-result pipeline.

Scientific verdict remains NEGATIVE. The stricter adjudication does not reverse a success; it prevents partial implementation from overstating how much of the locked protocol passed.

Controls, reproducibility, and limitations

Executed checks

- Valid split unit-overlap assertion returned zero.
- A duplicate-injection test detected ten deliberately overlapping units.
- A within-domain permutation association test returned $p = 0.00019996$ under the implemented null.
- Rerunning the deterministic final-gate calculator reproduced semantic JSON exactly.
- Processed data and deliverable files have SHA-256 manifests.

Unfinished or weaker-than-planned checks

- The frozen label-shuffle control required refitting rigorous models and showing no above-chance performance. It was not performed.
- The random-group control was represented by permuting optimism within domain, not by rebuilding random group assignments and rerunning models.
- The held-out-only batch-indicator negative control was not performed.
- The "clean rerun" did not start from downloaded archives in a fresh environment.
- G2's Holm-adjusted inferential contrasts were not implemented.
- The audit configuration table contains six diagnostics. The separately named group/cohort-delta dimension was effectively the target optimism quantity and was not included as an input, avoiding direct target leakage but diverging from the seven-dimension description.

Statistical limitations

The effective sample size is small at the domain level: three domains, with many configurations nested inside them. Configuration-level association can therefore look precise while domain transport is weak. Breast study folds and Parkinson subject folds also have different

biological meanings. No causal claim can be made that any audit dimension produces optimism. The analysis identifies warning patterns, not mechanisms.

Data and model limitations

Only one clinical tabular classifier, one transcriptomic classifier, and two longitudinal regressors were represented. Hist-gradient boosting and several planned preprocessing arms were not completed. Missingness, subgroup, calibration, and shift diagnostics were simplified proxies, not full stress-test batteries. The breast archive is curated secondary data; source-study assay and treatment heterogeneity remain. Diabetes is historical US inpatient data from 1999–2008. Parkinson telemonitoring has only 42 subjects.

Commercial and use-level translation

Intended user and decision

The credible near-term user is an ML validation lead, model-risk committee, journal reviewer, or regulated-product quality team deciding whether a reported performance estimate deserves external-validation work. The tool's role is to assemble evidence and flag missing safeguards. It must not decide that a model is safe, clinically effective, or market-ready.

Workflow stage	What the Round 1 artifact can do	What it cannot do
Study intake	Record independent unit, split logic, cohort boundaries, and missing diagnostics	Verify undocumented data provenance automatically
Pre-release review	Display naive-versus-group-aware deltas and diagnostic evidence	Provide a transportable pass/fail threshold
Monitoring	Define a repeatable dossier schema for shift and calibration checks	Replace post-deployment clinical monitoring
Audit	Preserve protocol, code, versions, checksums, and gate logic	Establish regulatory conformity or clinical utility

Product acceptance criteria for a future release

1. **Deterministic evidence schema:** every score must retain source columns, unit definition, split IDs, missing dimensions, and versioned bounds.
2. **Fail closed:** missing independent-unit IDs, corrupt input, fewer than five valid dimensions, or non-finite metrics must prevent a composite, not silently impute safety.
3. **Human-readable trace:** every flag must link to the exact rows and calculation that produced it.
4. **Domain-aware calibration:** a threshold must either have external-domain evidence or be explicitly labeled exploratory.
5. **Privacy:** local execution, data minimization, no raw patient export, access logs, retention controls, and aggregate-only reports by default.
6. **Security:** signed releases, dependency scanning, sandboxed parsers, input size limits, and reproducible builds.
7. **Monitoring and rollback:** drift, false-negative review, score-version comparison, and immediate reversion to manual-only status after performance regression.

Safety and regulatory posture

If used only as research quality assurance, the artifact is a decision-support checklist with quantitative aids. If marketed as determining clinical-model safety, regulatory acceptability, or patient risk, it may become part of a regulated quality or medical-device workflow and would require formal quality management, validated software lifecycle controls, cybersecurity, usability studies, and jurisdiction-specific review. Round 1 supplies none of those approvals. It should therefore carry a permanent "research and validation support only" label.

Economics

Round 1 does not contain measured labor savings, avoided validation failures, deployment costs, or willingness-to-pay data. No ROI claim is supportable. A commercial pilot should measure reviewer time per dossier, defect-detection yield, false reassurance rate, integration cost, and the cost of reproducing each flagged issue. Revenue or cost-savings estimates should wait for those observations.

Round 2: frozen-before-use development plan

Round 2 should answer why transport failed, not tune Round 1 until it passes. The following plan is derived from the observed G3/G4 failures and must be frozen before new outcomes are inspected.

1. **Separate evidence from scoring.** Retain the full six-dimension dossier, but compare three preregistered detectors: equal-weight composite, a simplified score excluding prevalence and calibration, and a rule-based “any severe hazard” detector. Round 1 labels may motivate these candidates but may not estimate their final weights.
2. **Add independent domains and configurations.** Target at least three additional open biomedical domains, including imaging or signal data and another multi-site clinical task. Within each domain, include multiple model families and fully executed preprocessing arms.
3. **Harmonize outcome scale.** Compare bounded percentile optimism, standardized loss difference, and domain-specific materiality thresholds. Freeze one primary mapping using simulations and theory, not Round 2 outcomes.
4. **Implement unit-level uncertainty.** Bootstrap patients, studies, and subjects at the raw prediction level; compute paired contrasts and Holm adjustment.
5. **Complete controls.** Refit label-shuffle, random-group, held-out batch-indicator, overlap, and duplicate-injection controls exactly as written.
6. **Prospective threshold evaluation.** Freeze scores and thresholds on a development set of domains, then evaluate once on untouched domains. Report calibration, sensitivity, specificity, false-negative cases, and decision curves.
7. **Product test.** Build a CLI only after the evidence schema is fixed. Test corrupt input, missing dimensions, runtime, peak memory, deterministic output, provenance trace, and report readability.
8. **Clean reproduction.** Rebuild from source archives in a pinned container and compare raw predictions, not only summary JSON.

Round 2 stop rule: if no candidate reaches the prespecified held-domain sensitivity and specificity without material subgroup or domain failure, stop universal-score development and publish the evidence-dossier approach as a non-scoring checklist.

Reproducibility record

The companion archive contains the frozen protocol and checksum, source and literature ledgers, environment ledger, all analysis scripts, processed result tables, log files, figures in PNG and SVG, conservative protocol adjudication, this HTML source, and the rendered PDF. Large source and processed input matrices are not embedded; their acquisition locations and SHA-256 hashes are provided to keep the artifact distributable. The archive itself has a separate SHA-256 checksum.

Primary public sources

- UCI Diabetes 130-US Hospitals: archive.ics.uci.edu/dataset/296
- UCI Parkinson Telemonitoring: archive.ics.uci.edu/dataset/189
- curatedBreastData archive: zenodo.org/records/21415886
- Nature Methods leakage guidance: [doi source](#)
- TRIPOD+AI: [BMJ 385:e078378](#)
- PROBAST+AI: [BMJ 388:e082505](#)
- FDA Good Machine Learning Practice: [FDA guidance page](#)

All numerical claims in this report are computed from the companion CSV/JSON files. Source URLs and fetched status are retained in sources.md. This document reports a negative result and should be cited with its protocol hash and archive checksum.